Shihaab Alam

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Profile

Second-year Computer Science student building systems across reinforcement learning, data analysis and desktop software, with extensive Python experience. Seeking placement or internship roles in software engineering, machine learning or data science.

Education

Aston University Birmingham, UK
BSc Computer Science (Sandwich Degree with Placement Year) *Sep. 2025 – Jun. 2029*

- Year 1: achieved a 2:1 overall, including 100% in the core AI & Data Science module
- Year 2 (2026–27): Data Structures, Algorithms & OOP; Software Engineering; Team Project; Programming Language Concepts

Projects

Celeste-RL | *Python, PyTorch, Stable-Baselines3 (PPO), Gymnasium* 2026

- Built a lockstep reinforcement-learning environment for *Celeste*, achieving ∼2,600 simulation FPS, ∼85× faster than the previous Python interface, with frame-for-frame playback validation
- Investigated a failed no-demonstration PPO campaign across 6M transitions and 3 seeds, identifying weak advantage signal among unsuccessful trajectories despite accurate value-function fitting
- Improved held-out Room 1 success from 52.2% to 74.8% (+22.6pp, p = .031) using varied-start, demonstration-assisted PPO under a predeclared evaluation protocol
- Engineered the system for unattended experiments with a 4.85M-step soak test, fault-injection testing and 469 unit tests, including diagnosing and fixing a reset-time memory leak that could consume ∼8 GB in 4 minutes

WCA Database Analysis | *Python, pandas, NumPy, SciPy, Jupyter, pytest* 2026

- Analysed a 1.5GB (6.4M row) public dataset; overturned a naive "home country advantage" hypothesis using paired statistical testing (p = 1.3e-67, n = 13,869), exposing selection bias
- Validated a trend-extrapolation model ($R^2 = 0.95$) against held-out data, uncovering a systematic 2.5–4 year forecast bias despite high in-sample fit
- Found and patched two data-integrity bugs in the source dataset, including a dtype mismatch silently corrupting roughly 40% of filtered rows, caught by a self-authored pytest suite automated via GitHub Actions CI

DualSense PC Overlay | *Python, PySide6, pydualsense, Spotipy, WinRT* 2026

- Built a PS5-style Windows overlay with live Spotify *Now Playing*, system controls (volume/power), and a controller-driven window switcher; alpha-tested by external users
- Fixed an input-lag bug – caused by OS thread starvation during CPU-heavy games – by elevating the controller-polling thread to highest priority
- Read the DualSense over raw USB HID (not XInput) at 100Hz on a background thread, routing events to the Qt main thread via signals and running Spotify Web API (PKCE OAuth) calls asynchronously to keep the UI responsive

Technical Skills

- **Languages**: Python, SQL, JavaScript, HTML/CSS
- **ML & Data**: PyTorch, Stable-Baselines3, Gymnasium, pandas, NumPy, SciPy, matplotlib
- **Frameworks & Libraries**: PySide6, Spotipy
- **Developer Tools**: Git, VS Code, Jupyter, pytest, GitHub Actions, PyInstaller
- **Other**: Linux (Ubuntu), Windows API / WinRT, Fusion 360

Extracurricular

Sustainable Waste Management System Design | *Engineers Without Borders* May 2026

- Designed a communal bokashi composting system for Ladywood, Birmingham (∼13,000 households, ∼400 hubs, est. £60k–£100k), chosen from five options via a weighted cost/social/environmental matrix
- Coordinated a multi-person team, including Fusion 360 CAD modelling, and pitched the final proposal as lead presenter